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PROFESSIONAL SUMMARY

Robotics and Automation undergraduate (CGPA 9.10/10.0) at Madras Institute of Technology with hands-on experience in embedded systems, robotics, and AI-driven applications. Skilled in Arduino-based system design, sensor interfacing, and rapid prototyping. Areas of interest include Computer Vision, industrial automation, embedded intelligence, and intelligent systems.

EDUCATION

B.E., Robotics and Automation - Madras Institute of Technology

2024 - 2028

- CGPA: 9.10 / 10.0 (current)
- Higher Secondary Certificate (2024): 94.16 percent
- Secondary School Certificate (2022): 95.2 percent
- JEE (Main) 2024: 81.82 percentile (CRL: 269,607; OBC-NCL: 94,238)

INTERNSHIPS & TRAINING

Computer Vision and AI Intern - PixelBotix

June 2026

- Completed hands-on training in image processing, object detection, and AI-based visual recognition; developed and deployed small-scale computer vision and AI projects.

Industrial Robotics and Automation Training - Birla Carbon and KUKA

February 28, 2026

- Trained on industrial robotic arm programming, automation workflows, and safety practices; exposed to Industry 4.0 concepts and KUKA robot operation.

PROJECTS

AgileX Scout Mini ROS 2 Jazzy Integration

ROS 2 Jazzy, C++, Python, Gazebo, RViz2

- Developed and integrated ROS 2 Jazzy driver nodes for the AgileX Scout Mini, handling real-time telemetry, cmd_vel multiplexing, and odometry tracking alongside Gazebo and RViz2 simulations.

Student Life Concierge AI Agent

Google ADK, Gemini API, FastAPI, Python, Pytest, Antigravity CLI

- Built a multi-agent FastAPI backend with prompt-injection defenses and a Pytest mock-service suite to automate student scheduling using Google ADK, Gemini API, and Antigravity CLI, developed during the Kaggle and Google 5-Day AI Agents Intensive.

Single-Axis Gimbal Stabilization System

Arduino Nano, MPU-6050, SG90 Servo

- Built a closed-loop PID gimbal stabilizer using an MPU-6050 IMU and embedded C on Arduino Nano for real-time angular correction.

Grip Stability and Stress Analyzer

Arduino Uno, HX711, Load Cell

- Built a load-cell measurement system with HX711 ADC to capture grip force and tremor patterns for stress-level estimation.

Adaptive Self-Mutating Authentication System

8051 (AT89S52), Embedded C

- Designed an 8051-based digital lock that regenerates a non-deterministic password at each login, resisting replay attacks and shoulder-surfing.

Mechanical Rotary Device

SolidWorks, Design Patent Pending

- Designed a dual-mode Geneva wheel and dog clutch rotary mechanism in SolidWorks; filed for design patent.

HACKATHONS & COMPETITIONS

Smart India Hackathon 2025 - Finalist

Delhi, 2025

- Collaborated on reverse-engineering multi-architecture firmware binaries to identify cryptographic primitives and protocols using AI and machine learning.

H2TS Hackathon 2025 - 2nd Place

2025

- Collaborated with a cross-functional team to build and deploy an AI-driven SOS platform for rapid emergency response.

TECHNICAL SKILLS

- **Embedded Systems:** Arduino Uno R3, Arduino Nano, 8051 (AT89S52), Sensor Interfacing, Hardware Timers, I2C/SPI Communication
- **Control Systems and Robotics:** PID Control, Closed-Loop Feedback, Motion Control, Inertial Sensing, Stabilization Mechanisms, ROS 2
- **Programming:** Python (Intermediate), Embedded C (Fundamentals), MATLAB/Simulink (Intermediate)
- **Operating Systems:** Linux (Ubuntu)
- **AI and Agentic Systems:** Google Agent Development Kit (ADK), Gemini API, Ollama, Llama 3.2, FastAPI, Antigravity CLI, Prompt Engineering
- **Computer Vision:** Image Processing, Object Detection, AI-based Visual Recognition (OpenCV-level coursework)
- **Design and EDA:** SolidWorks, KiCAD, Proteus, Canva
- **Cybersecurity / Reverse Engineering:** Binwalk, Ghidra, Firmware Binary Analysis (working knowledge of tools and workflow)

CERTIFICATIONS

- Automation in Manufacturing - IIT Guwahati, NPTEL (Elite, 67 percent)
- Foundations of Prompt Engineering - AWS Training and Certifications

LEADERSHIP & ACTIVITIES

- Placement Representative, 2026-Present - Liaison between students and placement coordinators.
- Joint Treasurer, MIT Quill, 2025-26 - Managed treasury for the college literary club.
- Office Bearer and Content Writer, SAHA, 2025-26 - Organized academic events for the Science and Humanities Association.
- Member, ARME (Association of Robotics and Mechatronics Engineers), 2024-Present.
- YRC Volunteer, MIT, 2024-Present - Organized Youth Fest events for visually challenged individuals.

LANGUAGES

English (Proficient) | Tamil (Proficient) | Telugu (Conversational) | Hindi (Elementary)